

A Critical Survey of Driver Distraction Detection Systems

RGB Image Classification, Low-Light Recognition and IR/NIR Robustness

Datasets, Evaluation Biases, Domain Shift and Deployment Evidence

Dulhara Anjana Langappuli

Department of Physics, Engineering and Computer Science

University of Hertfordshire, United Kingdom

Email: dl22aaw@herts.ac.uk

Abstract

Driver distraction detection is often presented as a solved image-classification problem because several studies report high accuracy on controlled RGB datasets. In reality, results depend on a variety of factors including dataset used, lighting, camera positioning, subject split, task definition, and evaluation protocol. The consideration is especially important in the case of in-cabin monitoring systems, which should be able to recognize distractions in daytime RGB as well as at nighttime, in low-light and NIR images.

In this survey, driver distraction detection is evaluated from the perspective of lighting robustness. RGB datasets (State Farm, AUC, LDDb, and DAD) are compared with low-light, infrared (IR), near-infrared (NIR), NoIR-camera, and multimodal driver monitoring datasets such as Low-Light Driver Distraction Dataset, DMD/VicomTECH, and 100-Driver. Techniques employed include handcrafted features, CNN-based image classification, lightweight CNN architectures, CNN-machine learning hybrids, object detection, temporal models, visible-to-infrared translation, IR-like synthesis, and hardware-friendly deployment. Instead of viewing accuracy as a performance leaderboard, the review focuses on whether the papers offer subject-wise splitting, cross-dataset evaluation, evaluation in low light, class-based metrics, and implementation. The result of analysis shows that the main issue of the field is not lack of classification models, but the lack of evidence for generalization to unseen subjects and lighting conditions. Thus, the best area for future research is lighting-robust driver distraction recognition, i.e., evaluation in RGB, low-light, and IR/NIR conditions instead of RGB accuracy claims.

Keywords: driver distraction detection; driver monitoring; low-light recognition; infrared; near-infrared; NoIR camera; image classification; domain adaptation; subject-wise split; synthetic IR-like augmentation.

1 Introduction

Driver distraction hampers the driver's capability to focus on the driving process and respond to the changing environment safely. The notion is usually characterized in terms of visual, manual and cognitive distraction. Visual distraction refers to when the driver takes his/her eyes off the road, manual distraction – when the driver's hands are off the steering wheel, and cognitive distraction implies that the driver's mind is off the task of driving (Regan et al. 2011; Kashevnik et al. 2021). This classification is convenient, but in practice there might be overlap. Texting while driving, for example, may imply the eyes, hands and attention simultaneously.

The importance of the topic lies in the direct link between driver distraction and road safety. According to the National Highway Traffic Safety Administration, in 2022, distracted driving accounted for 3,308 deaths and 289,310 injuries (National Highway Traffic Safety Administration 2024). Vehicle regulations have been developing in the direction of active monitoring of driver distractions, including drowsiness, attention, distraction warning (European Commission 2024). Driver distraction detection is thus not only an image classification computer vision task; it is an in-cabin perception task crucial for safety.

Camera-based approaches are extensively researched due to their ability to detect visible driver behaviors without any sensors on the driver's person. Driver-facing camera allows detecting such behaviors as mobile phone use, texting, drinking, reaching behind, radio manipulation, and conversing with the passenger. Due to that, most of the research on driver distraction detection formulates the problem as an image classification task. The State Farm Distracted Driver Detection dataset is a typical example in this case since it includes labels for ten driver behavior classes on an image level (Kaggle 2016). Yet, being accurate on the daytime RGB benchmark is not sufficient to address driver distraction detection as an application.

The major problem with much of the existing literature is the gap between the dataset performance and the practical setting of operation. One of the clear examples of this gap is the lighting condition. It can vary drastically in terms of time of the day (daylight, night), lighting (street light, tunnel lighting, shadows, and infrared). RGB-only datasets are incomplete in that regard. Thus, driver monitoring in low-light, IR, NIR and NoIR settings are not secondary topics; they are central to the applicability of the driver distraction detector.

This survey focuses on the latter. From the image classification to the low-light and IR/NIR robustness, this review will consider evaluation bias and deployment issues. There are five questions that the paper is aimed to answer:

1. Which datasets support RGB, low-light, IR, NIR and multimodal driver distraction or driver monitoring research?
2. Which task formulations are being used and where do image classification, object detection and temporal recognition differ?
3. How should reported accuracy be interpreted when datasets, splits and lighting conditions differ?
4. Which studies provide evidence of robustness across unseen drivers, lighting domains or sensor modalities?
5. What experimental direction would make future driver distraction detection work more convincing?

The contribution of this survey is a lighting-robustness-centred reading of the literature. Unlike a broad catalogue of driver monitoring methods, the review separates RGB classification evidence from low-light and IR/NIR evidence and it assesses whether reported results are supported by fair evaluation protocols.

2 Review Methodology

2.1 Search Strategy

A PRISMA-style process was used to organise the search, screening, eligibility assessment and final selection of studies. PRISMA was used as a reporting structure because it makes the selection process more transparent (PRISMA 2020; Page et al. 2021). The search was carried out in June 2026 using IEEE Xplore, Google Scholar, SpringerLink, MDPI, Elsevier/ScienceDirect-related sources, arXiv, dataset repositories and citation chasing.

Three groups of search phrases were used:

- **RGB driver distraction classification:** “driver distraction detection”, “distracted driver detection image classification”, “State Farm distracted driver detection”, “driver behaviour recognition deep learning”, “driver distraction CNN MobileNet”.
- **Detection and deployment:** “driver distraction detection YOLO”, “real-time driver distraction detection”, “driver monitoring system edge device”, “driver distraction FPGA”, “embedded driver monitoring inference”.
- **Lighting robustness:** “driver distraction low light dataset”, “night-time driver distraction detection”, “infrared driver distraction detection”, “near infrared driver monitoring”, “NoIR camera driver distraction”, “visible infrared image translation driver distraction”, “RGB NIR driver monitoring dataset”.

Most selected papers were published from 2020 onwards. Older studies were retained only where they provided useful definitions, early benchmarks, survey context or baseline methods.

2.2 Search Sources and Record Counts

Table 1: Search sources and initial records identified.

Source	Records	Main focus of results
IEEE Xplore	32	Driver distraction, CNNs, FPGA, low-light NoIR and IR/NIR studies.
Google Scholar	24	Citation search, arXiv papers, dataset papers and review studies.
SpringerLink	8	Driver monitoring, edge/cloud systems and deep learning studies.
MDPI	7	Sensors, applied sciences, driver state monitoring and multimodal recognition.
Elsevier / ScienceDirect-related sources	8	Intelligent systems, driver behaviour and accident-related monitoring studies.
arXiv and preprint sources	5	DMD, edge AI, synthetic data and emerging methods.
Dataset repositories and citation tracking	6	Mendeley low-light dataset, DMD website and Zenodo dataset records.
Total identified	90	Records before duplicate removal.

2.3 Inclusion and Exclusion Criteria

Table 2: Inclusion and exclusion criteria.

Inclusion criteria	Exclusion criteria
Studies on driver distraction detection, driver behaviour recognition or driver monitoring.	Studies unrelated to driver monitoring, in-cabin sensing or distraction-related behaviour.
Studies using RGB images, video, low-light images, NoIR, IR, NIR, depth, thermal or multimodal sensors.	Studies using only non-visual signals with no clear link to camera-based monitoring.
Studies using CNNs, lightweight networks, object detectors, SVM, PCA, LSTM/GRU, attention, transformers, translation or hybrid methods.	Papers without clear dataset, method, evaluation result or technical contribution.
Studies reporting task type, dataset, participant information, camera setup, metrics, split protocol or deployment information.	Duplicate papers, inaccessible papers, blog posts, incomplete documents or non-academic material.
Studies relevant to low-light, night-time, IR/NIR, NoIR or day/night driver monitoring.	General low-light image enhancement studies without driver monitoring relevance.

2.4 Study Selection Process

The selection process is shown in Figure 1. The initial search identified 90 records. After removing 22 duplicates, 68 records were screened by title and abstract. Twenty-nine records were excluded at this stage because they were outside the scope, too general, or technically incomplete. Thirty-nine full-text papers were assessed. Eight papers were excluded after full-text assessment. Thirty-one papers remained eligible and 20 core studies were selected for detailed comparison. The remaining 11 eligible papers were retained as supporting literature.

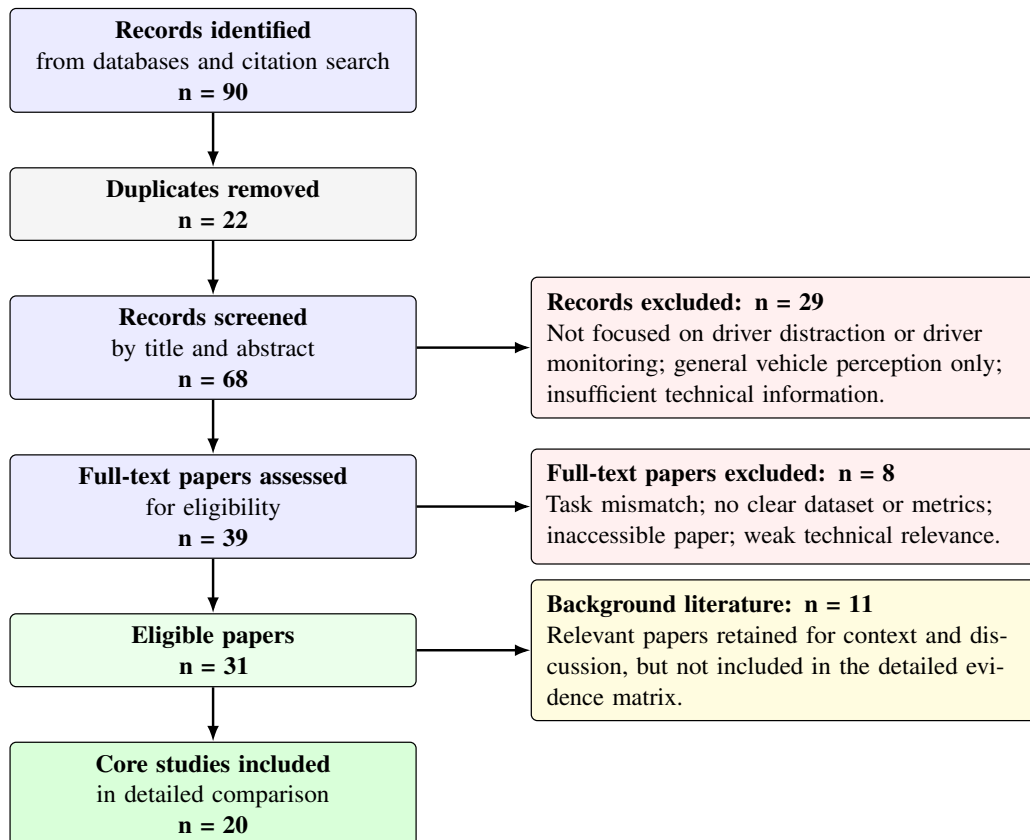


Figure 1: PRISMA-style selection process used for the survey.

2.5 Reasons for Exclusion

Table 3: Full-text exclusion and background-use reasons.

Reason	Number	Decision
Full text did not provide clear dataset, method or metric information.	3	Excluded after full-text assessment.
Task was too far from driver distraction or driver monitoring.	2	Excluded after full-text assessment.
Paper focused mainly on drowsiness/fatigue without useful distraction relevance.	1	Excluded after full-text assessment.
Paper was inaccessible or incomplete.	1	Excluded after full-text assessment.
Duplicate or highly similar technical contribution.	1	Excluded after full-text assessment.
Relevant but less central to the evidence matrix.	11	Retained for background discussion.

3 Research Landscape

Driver distraction detection connects several research layers: distraction psychology, sensing modality, computer vision task, model family, dataset type, evaluation protocol and deployment platform. Confusion between these layers is one reason why the literature can appear more consistent than it actually is. For example, a ten-class image classifier trained on State Farm, a YOLO-based phone detector, a landmark-based drowsiness monitor and an IR-based night-time driver monitoring system are all relevant, but they do not solve the same task.

Figure 2 summarises the landscape. The highlighted route is the main focus of this survey: RGB driver-distraction classification, low-light/IR/NIR data, domain adaptation and fair evaluation.

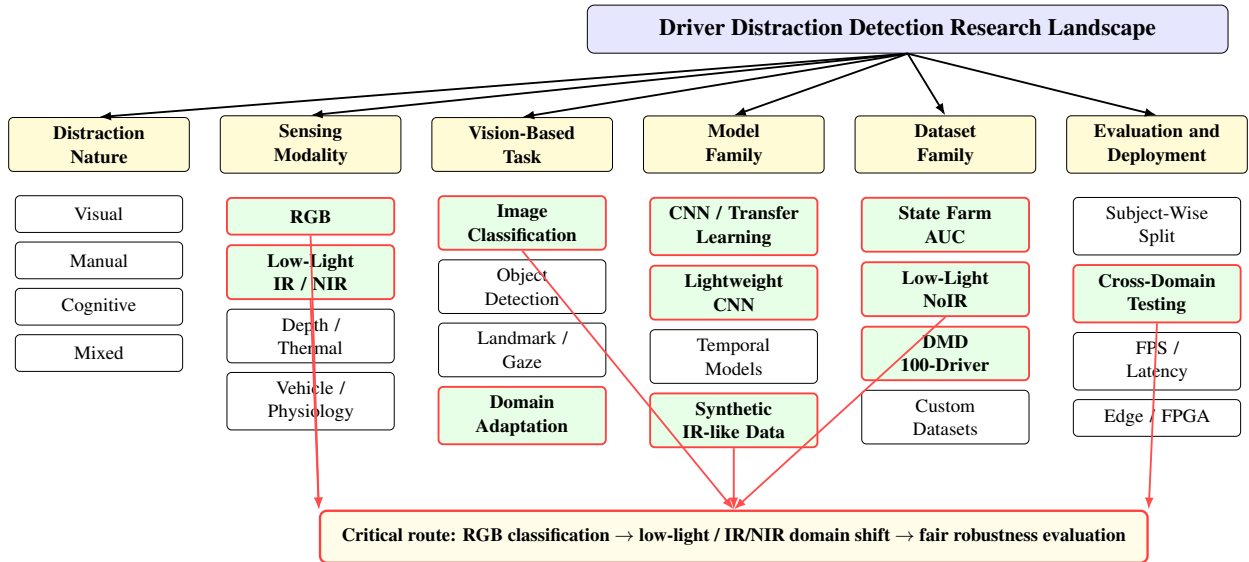


Figure 2: Research landscape of driver distraction detection, with emphasis on the link between RGB classification, low-light/IR/NIR domain shift and fair robustness evaluation.

Table 4: Survey taxonomy used to organise the literature.

Layer	Branches	Why it matters
Distraction nature	Visual, manual, cognitive, mixed	Defines the behaviour being recognised.
Sensing modality	RGB, NoIR, IR/NIR, depth, thermal, vehicle signals	Determines whether the system can work across lighting conditions.
Task type	Image classification, detection, landmarks, temporal recognition, domain adaptation	Determines label type and whether results are comparable.
Dataset family	State Farm, AUC, LDDDB, DAD, DMD, low-light NoIR, 100-Driver	Determines subject diversity, lighting range and generalisation risk.
Model family	CNN, lightweight CNN, CNN-ML, YOLO, LSTM/GRU, transformer, translation	Determines modelling strategy and computational burden.
Evaluation	Accuracy, macro-F1, per-class recall, confusion matrix, subject-wise split, cross-domain test	Determines whether reported performance is reliable.
Deployment	CPU, GPU, edge device, Raspberry Pi, Jetson, FPGA	Determines whether the method is only offline or practically deployable.

4 Datasets and Lighting Conditions

Datasets are not neutral containers of images. In driver monitoring, the dataset often determines the apparent difficulty of the problem. The same model can look strong on a controlled RGB dataset and weak under different lighting, camera placement or driver population. This section therefore separates RGB classification datasets from low-light and IR/NIR datasets.

4.1 RGB Image-Classification Datasets

The State Farm Distracted Driver Detection dataset is widely used for image-level classification. It contains ten classes: safe driving, texting right, talking on the phone right, texting left, talking on the phone left, operating the radio, drinking, reaching behind, hair/makeup and talking to a passenger (Kaggle 2016). Its value is that the task is simple to formulate and easy to reproduce: one image, one label.

The weakness is that it does not represent all deployment conditions. It is not an object-detection dataset because it does not provide bounding boxes. It is also vulnerable to subject leakage if images are randomly split. If the same driver appears in training and testing, the model may learn identity, clothing, cabin background or camera position. For this reason, subject-wise splitting is more meaningful than random image splitting.

Other RGB datasets, including AUC, LDDDB, DAD and custom datasets, provide additional benchmarks (Eraqi et al. 2019; Li et al. 2022; Wu et al. 2024). They differ in class labels, driver diversity, camera angle and evaluation protocol, so their reported metrics should not be treated as directly interchangeable.

4.2 Low-Light, IR and NIR Datasets

Low-light and IR/NIR studies are important because practical driver monitoring must work when visible light is poor. Table 5 compares the most relevant low-light and IR/NIR studies identified in this survey.

Table 5: Low-light, IR/NIR and day-night datasets relevant to driver monitoring.

Study / dataset	Year	Sensor / modality	Scale	Task relevance	Main caution
Saad et al. low-light dataset	2020	Raspberry Pi NoIR camera with IR LEDs	70 drivers, 5 car models, 52,350 frames, 10 classes	Direct low-light driver distraction recognition	Split protocol and cross-domain transfer need careful checking.
Mendeley low-light dataset	2020	NoIR low-light support	Dataset associated with Saad et al.	Useful for night-time recognition	Should not be treated as equivalent to real NIR fleet data.
Shen and Lin	2023	Visible and infrared images	Day/night visible-infrared translation study	Directly relevant to RGB-IR domain shift	Synthetic translation is not physical IR recovery.
DMD / VicomTECH	2020	RGB, depth, IR videos	41 hours, 37 drivers, 3 camera views	Multimodal driver monitoring and distraction activities	Broader than distraction classification only.
100-Driver	2023	Daytime RGB and night-time NIR	100 drivers, 470,208 images, 4 views, 5 vehicles	Strong day/night and cross-camera evidence	Task categories differ from State Farm.
InSight	2020	NIR LED-based low-light setup	15 drivers, controlled and real-road low-light settings	Low-light driver state monitoring	Not a full 10-class distraction dataset.
IR driver monitoring	2023	Infrared video	Real-time driver monitoring	Supports night-time in-cabin monitoring	Focuses more on drowsiness/head pose than distraction.
RGBDT activity recognition	2024	RGB, depth and thermal infrared	Multimodal activity recognition	Useful for lighting-robust activity recognition	Sensor setup differs from normal RGB cameras.

Saad et al. are useful because the study directly addresses low-light driver distraction recognition and reports a best accuracy of 95.48% using Convolutional GRU (Saad et al. 2020). The result is important, but it should not be read as proof that low-light distraction recognition is solved. A stronger interpretation is that NoIR-supported low-light data can support high within-dataset performance, while the harder question is whether models trained under one lighting and camera setup generalise to different drivers, vehicles, cameras and night-time domains. DMD/VicomTECH and 100-Driver broaden the discussion. DMD provides RGB, depth and IR data from multiple in-cabin cameras and includes distraction-related activities (Ortega et al. 2020). The 100-Driver dataset adds a larger driver population and includes daytime RGB and night-time NIR modalities (Wang et al. 2023). These datasets suggest that the next useful step for the field is not another RGB-only accuracy comparison, but controlled testing across lighting and sensor domains.

4.3 What the Dataset Evidence Shows

Table 6 summarises the dataset evidence more directly. The table is not intended to rank datasets; it shows which kinds of evidence are available and which are still missing.

Table 6: Dataset evidence for lighting-robust driver distraction recognition.

Dataset family	RGB	Low-light / IR	Driver diversity	Main value for this survey
State Farm-style datasets	Strong	Weak	Moderate	Useful baseline for image classification, but limited lighting robustness evidence.
AUC / posture datasets	Strong	Limited	Moderate	Useful for posture and distraction classification comparisons.
Low-light NoIR dataset	Limited RGB comparison	Strong	70 drivers	Directly relevant to low-light distraction recognition.
DMD / VicomTECH	Strong	Strong	37 drivers	Useful multimodal evidence across RGB, depth and IR.
100-Driver	Strong	Strong NIR	100 drivers	Strong evidence for day/night and cross-view evaluation.
Custom real-time datasets	Variable	Variable	Often small	Useful for deployment examples, but reproducibility is often weaker.

5 Algorithm Families

The reviewed studies use several model families. The important point is not simply which model reports the highest accuracy, but whether the model is evaluated under a fair and realistic protocol.

Handcrafted features and traditional machine learning. Earlier methods used HOG, SIFT, contour descriptors, skin segmentation or gaze-related features with SVM or random forest classifiers (Hssayeni et al. 2017; Zhao et al. 2012). These methods are easier to interpret but depend heavily on manually designed features.

CNN image classification. CNN-based classification remains the natural baseline for image-level datasets such as State Farm. ResNet, VGG, EfficientNet, MobileNet, Swin Transformer and custom CNNs have all been used for driver behaviour classification (Huang et al. 2020; Li et al. 2022; Khan et al. 2023; Zhang et al. 2024). The risk is that CNNs can learn dataset-specific cues if the split protocol is weak.

Lightweight CNNs. Lightweight models are important for practical deployment because in-cabin systems may need to run on limited hardware. MobileNet-style models and compact architectures are therefore useful, but they should be evaluated using model size, FPS and latency, not only accuracy (Wu et al. 2024; Wang and Wu 2022).

Hybrid CNN-machine learning pipelines. Some studies use CNNs for feature extraction and then apply PCA, SVM or other classifiers (Revathi et al. 2025; Alkinani et al. 2022). This can be useful for dimensionality reduction and controlled comparison, but it does not remove the need for subject-wise testing.

Object detection. YOLO, Faster R-CNN and EfficientDet are relevant when the target is to localise phones, hands or objects (Sajid et al. 2021; Kanimozhi et al. 2025; Elshamy et al. 2024). These methods should not be directly compared with image classification unless the dataset contains compatible bounding-box annotations.

Temporal models. Driver behaviour is often temporal. A single image may not capture the difference between reaching, turning, checking a mirror or preparing to interact with a device. Saad et al. used temporal models such as ConvLSTM and ConvGRU on low-light data, with Convolutional GRU reporting the best result (Saad et al. 2020). This supports sequence modelling, especially when frame order is available.

Visible-infrared translation and synthetic IR-like generation. Visible-infrared translation is relevant because daytime RGB and night-time IR/NIR belong to different domains (Shen and Lin 2023). The wording must be precise: RGB images cannot be transformed into true physical IR images. A more defensible aim is to generate synthetic IR-like or NIR-style samples for augmentation and domain adaptation, then test whether they improve robustness on real low-light or IR/NIR data.

6 Deployment Evidence

Deployment evidence is weaker than classification evidence. Many studies report accuracy but do not report latency, FPS, power, model size or hardware resource usage.

CPU and GPU. GPU results are useful for training and offline evaluation, but they do not prove in-vehicle deployment. A model that performs well on a GPU may be too slow or power-hungry for an embedded platform.

Edge devices. Raspberry Pi and Jetson-style platforms appear in several real-time monitoring studies (Kurian and Soori 2023; Mantrala et al. 2025; Hariharan et al. 2023). These studies are useful because they move closer to deployment, but they often use smaller datasets or custom test settings.

FPGA. FPGA work is relevant to low-latency and energy-efficient inference, but the evidence must be hardware-specific. A useful FPGA paper should report quantisation, throughput, latency, FPS, power, energy per frame, LUTs, DSPs, BRAM and accuracy loss after compression or mapping (Heenetimulla et al. 2022). Without these details, the hardware contribution is too vague.

7 Evidence Matrix of Core Studies

The detailed comparison is divided into two readable tables. Table 7 gives the technical facts. Table 8 gives the contribution and limitation.

Table 7: Technical facts of the core studies.

No.	Authors / year	Dataset	Task type	Method	Metric
1	Kashevnik et al. (2021)	Review-based	Survey	Literature review and framework	N/A
2	Revathi et al. (2025)	State Farm	Image classification	CNN feature extraction, PCA, SVM	99.3% Acc.; 98.9% macro-F1
3	Saad et al. (2020)	Low-light NoIR dataset	Image / sequence classification	VGG16, ResNet50V2, InceptionV3, MobileNetV2, ConvLSTM, ConvGRU	95.48% Acc.
4	Shen and Lin (2023)	Visible and infrared driver data	Day/night recognition	Visible-infrared translation and recognition	Reported
5	Ortega et al. (2020)	DMD / Vi-comTECH	Driver monitoring	RGB-depth-IR multimodal dataset	N/A
6	Wang et al. (2023)	100-Driver	Driver distraction classification	RGB/NIR, cross-camera and cross-vehicle evaluation	Reported
7	Mantrala et al. (2025)	Mixed video/image sources	Hybrid detection	MediaPipe, YOLOv5, MobileNetV3, rules	88.1% Acc.; 0.86 F1
8	R. K. and Mathew (2024)	State Farm	Image classification	ReLU-BiLSTM with attention	95.87% Acc.; 95.24% F1
9	Kanimozhi et al. (2025)	Distracted-driver data	Object detection / classification	YOLO and CNN	88-92% reported
10	Heenetimulla et al. (2022)	Driver posture and heart-rate data	Mobile-use detection and health monitoring	FPGA, camera, smartwatch	93.49% mobile-use detection
11	Wei et al. (2022)	AUC dataset	Image classification	ResNet-50	43.5% ten-class; 77.9% binary
12	Kurian and Soori (2023)	Real-time camera data	Drowsiness / distraction monitoring	Haar, Dlib, EAR, MAR, gaze tracking	94.1% drowsiness; 89% distraction
13	Wu et al. (2024)	DAD dataset	Distraction recognition	DDDNet, MobileNetV2, contrastive learning	87.12% Acc.; 0.9009 AUC
14	Abouelnaga et al. (2018)	AUC	Posture classification	CNN ensemble	95.98% Acc.
15	Hssayeni et al. (2017)	Distracted-driver images	Classification	Handcrafted features vs. deep learning	Not reported
16	Huang et al. (2020)	State Farm	Image classification	Hybrid CNN framework	Reported
17	Li et al. (2022)	LDDb and State Farm	Image classification	Octave-like CNN	Reported
18	Sajid et al. (2021)	State Farm	Detection-style framework	EfficientDet, Faster R-CNN, YOLOv3	Reported
19	Khan et al. (2023)	State Farm / AUC	Image classification	EfficientNet with channel attention	99.58% reported
20	Zhang et al. (2024)	AUC-DD / State Farm	Image classification	Swin Transformer	99.71% reported

Table 8: Main contribution and limitation of the core studies.

No.	Study	Main contribution	Main limitation / caution
1	Kashevnik et al. (2021)	Provides a useful taxonomy and framework for driver distraction detection methods.	Review paper; does not evaluate a new model.
2	Revathi et al. (2025)	Strong State Farm result with accuracy and macro-F1.	High result still depends on split protocol and subject independence.
3	Saad et al. (2020)	Direct low-light driver distraction recognition using NoIR-supported data.	Needs careful interpretation of driver/session independence and cross-domain transfer.
4	Shen and Lin (2023)	Links daytime visible and night-time infrared domains.	Translation output should not be treated as true physical IR.
5	Ortega et al. (2020)	Provides broad RGB-depth-IR driver monitoring data.	Wider than distraction classification alone.
6	Wang et al. (2023)	Provides large-scale day/night RGB/NIR driver data.	Task labels differ from State-Farm-style classification.
7	Mantrala et al. (2025)	Demonstrates an edge-AI style pipeline.	Mixed sources make direct benchmarking difficult.
8	R. K. and Mathew (2024)	Shows a lightweight temporal-style model on State Farm.	Needs stronger unseen-driver and deployment evidence.
9	Kanimozhi et al. (2025)	Represents object-detection-based distraction work.	Detection and classification results are not directly comparable.
10	Heenetimulla et al. (2022)	Gives a hardware-related FPGA example.	Task differs from ten-class driver distraction classification.
11	Wei et al. (2022)	Shows binary safe/unsafe classification is easier than ten-class recognition.	Demonstrates difficulty of fine-grained classification.
12	Kurian and Soori (2023)	Gives a real-time Raspberry Pi monitoring example.	Stronger on drowsiness than full distraction recognition.
13	Wu et al. (2024)	Contributes lightweight/open-set recognition direction.	Dataset and task differ from State Farm.
14	Abouelnaga et al. (2018)	Important benchmark for posture classification.	Older and more controlled than deployment settings.
15	Hssayeni et al. (2017)	Motivates CNN features over handcrafted features.	Some metrics are not available in the current summary.
16	Huang et al. (2020)	Provides State-Farm-based hybrid CNN comparison.	Exact split and generalisation protocol should be checked.
17	Li et al. (2022)	Presents efficient CNN design for distraction detection.	Cross-dataset robustness needs careful interpretation.
18	Sajid et al. (2021)	Useful detection-style comparison using modern detectors.	Annotation compatibility must be considered.
19	Khan et al. (2023)	Reports strong EfficientNet-channel-attention performance.	High accuracy needs careful split-protocol checking.
20	Zhang et al. (2024)	Represents transformer-based distracted-driver classification.	Computational cost may be higher than lightweight alternatives.

8 Critical Quantitative Synthesis

A direct accuracy leaderboard is not appropriate for this topic. The studies use different datasets, lighting conditions, split protocols, metrics and task definitions. A 99% result on a random split can be less convincing than a lower score under subject-wise or cross-dataset testing. For that reason, Table 9 interprets reported results together with evaluation context.

Table 9: Reported performance values and interpretation context.

Study	Reported value	Dataset / domain	Evaluation concern	Interpretation
Revathi et al.	99.3% Acc.; 98.9% F1	State Farm RGB	Split details matter	Strong within-dataset result, but not proof of low-light robustness.
Saad et al.	95.48% Acc.	Low-light NoIR	Cross-domain generalisation unclear	Important low-light evidence, but should be tested against other domains.
Mantrala et al.	88.1% Acc.; 0.86 F1	Mixed sources	Dataset mixing	Useful system-level evidence, but not a clean benchmark comparison.
Wei et al.	43.5% ten-class; 77.9% binary	AUC	Task difficulty changes result	Shows binary detection is much easier than fine-grained classification.
Wu et al.	87.12% Acc.; 0.9009 AUC	DAD	Different task setting	Useful lightweight evidence, but not directly comparable with State Farm.
Khan et al.	99.58% reported	State Farm / AUC	Split and leakage risk	High result should be interpreted through protocol details.
Zhang et al.	99.71% reported	AUC-DD / State Farm	Computational cost and split protocol	Transformer result is strong, but deployment and fairness need checking.

Table 10: Evidence strength across major evaluation requirements. A tick means the study direction directly supports the evidence type; a dash means weak, indirect or not clearly reported evidence.

Study group	RGB	Low-light	IR/NIR	Subject split	Deployment	Main reading
State Farm CNN studies	✓	–	–	Mixed	–	Strong for RGB classification, weaker for lighting robustness.
Low-light NoIR study	–	✓	✓	Needs checking	–	Key evidence for low-light distraction recognition.
DMD / 100-Driver studies	✓	✓	✓	Stronger	Mixed	Best evidence for modality and lighting diversity.
YOLO / object detection studies	✓	Mixed	Mixed	Mixed	Mixed	Useful for localisation, but annotation type differs.
Edge / FPGA studies	Mixed	Mixed	Mixed	Often weak	✓	Useful for deployment, but not always strong in dataset evaluation.

The main message from these tables is that high reported accuracy is not enough. The more useful question is whether the study tests the model under a realistic source of variation: unseen drivers, different lighting, different camera views, or a different dataset.

9 Key Trends

Three trends are clear from the reviewed literature.

First, RGB image classification remains the easiest entry point into driver distraction detection. Datasets such as State Farm allow clean training and evaluation and CNN-based methods can achieve high reported accuracy. The weakness is that RGB benchmarks alone do not establish robustness.

Second, low-light and IR/NIR driver monitoring is becoming more important. Saad et al., DMD/VicomTECH and 100-Driver show that the literature is moving towards NoIR, IR, NIR and multimodal data (Saad et al. 2020; Ortega et al. 2020; Wang et al. 2023). This gives the field a stronger direction than repeating RGB-only comparisons.

Third, many papers still report incomplete evidence. Accuracy is commonly reported, but subject-wise splits, cross-dataset testing, false-negative analysis, model size, latency and FPS are less consistent. This matters because a driver monitoring system must generalise, not only classify a familiar benchmark.

10 Open Challenges

Lighting robustness. A reliable in-cabin system must work across daylight, night-time, shadow, tunnel and artificial illumination. The field still lacks enough studies that train on one lighting domain and test on another.

Subject leakage. If the same driver appears in both training and test sets, results can be inflated. Subject-wise splits are essential where subject labels are available.

Dataset bias. Datasets differ in camera position, vehicle interior, driver demographics, lighting, class labels and annotation style. Cross-dataset testing is needed to reveal whether a model has learned behaviour or dataset-specific appearance.

Task mismatch. Image classification and object detection are different problems. Image classification uses image-level labels, while detection requires bounding boxes. They should not be compared as if they used the same evidence.

Synthetic IR-like data. Synthetic IR-like or NIR-style augmentation is promising, but the claim must be limited. It can support robustness testing and domain adaptation, but it does not recreate the physical information captured by a real IR sensor.

Deployment evidence. Deployment claims should include latency, FPS, memory, power, model size and hardware resource usage. Without these metrics, a method remains mainly an offline model.

11 Future Research Direction

The most useful next research direction is not another general comparison of CNNs on RGB images. A stronger direction is a controlled study of lighting-domain robustness in driver distraction detection.

A practical experimental design would compare:

1. RGB-only training and RGB testing.

2. RGB training with conventional augmentation.
3. RGB training with synthetic IR-like or NIR-style augmentation.
4. Testing on subject-wise RGB splits.
5. Testing on real low-light, NoIR, IR or NIR data where available.
6. Reporting accuracy, macro-F1, per-class recall, false negatives and confusion matrices.

The key comparison is not simply whether synthetic IR-like images improve average accuracy. The more important question is whether they reduce failure under lighting shift and whether they improve difficult distraction classes without increasing false negatives in safety-critical cases.

Hardware acceleration can be treated as a second direction after the AI pipeline is stable. A convincing hardware study should report model compression, quantisation, latency, throughput, FPS, energy per frame, power, LUT usage, DSP usage, BRAM usage and accuracy loss after hardware mapping.

12 Threats to Validity

This survey has limitations. Some industrial driver monitoring datasets and systems are not publicly accessible. Several papers report results without full details of split protocol or deployment hardware. Low-light and IR/NIR studies do not always use the same task definition as State-Farm-style image classification. Reported performance values are therefore interpreted as evidence within context, not as a universal ranking.

13 Conclusion

This survey reviewed driver distraction detection with a focus on the transition from RGB image classification to low-light and IR/NIR robustness. The review shows that the field has many strong classification models, but less convincing evidence of robustness under lighting shift, unseen drivers and different sensor domains. RGB benchmarks such as State Farm remain useful, but they should not be treated as sufficient evidence for practical driver monitoring.

The main research opportunity is lighting-robust driver distraction recognition. Future work should combine RGB, low-light and IR/NIR evidence with subject-wise testing, cross-domain evaluation and class-level error analysis. Synthetic IR-like or NIR-style augmentation is a promising direction if it is framed carefully as domain adaptation rather than true IR recovery. Deployment and FPGA acceleration are also valuable, but they require separate hardware-specific evidence.

References

- [1] Alkinani, M. H., W. Z. Khan, and Q. Arshad (2022). “HSDDD: A Hybrid Scheme for the Detection of Distracted Driving through Fusion of Deep Learning and Handcrafted Features”. In: *Sensors* 22.5, p. 1864. DOI: [10.3390/s22051864](https://doi.org/10.3390/s22051864).
- [2] Barr, L., H. Howarth, S. Popkin, and R. J. Carroll (2005). *A Review and Evaluation of Emerging Driver Fatigue Detection Measures and Technologies*. Tech. rep. Washington, DC: Federal Motor Carrier Safety Administration.
- [3] Dong, Y., Z. Hu, K. Uchimura, and N. Murayama (2011). “Driver inattention monitoring system for intelligent vehicles: A review”. In: *IEEE Transactions on Intelligent Transportation Systems* 12.2, pp. 596–614. DOI: [10.1109/TITS.2010.2092770](https://doi.org/10.1109/TITS.2010.2092770).
- [4] Dontoh, A., S. Ivey, L. Sirbaugh, and A. Aboah (2025). *A Review Paper of the Effects of Distinct Modalities and ML Techniques to Distracted Driving Detection*. arXiv: [2501.11758](https://arxiv.org/abs/2501.11758) [cs.CV]. URL: <https://arxiv.org/abs/2501.11758> (visited on 06/22/2026).
- [5] Dontoh, A., S. Ivey, L. Sirbaugh, A. Danyo, and A. Aboah (2025). *Visual Dominance and Emerging Multimodal Approaches in Distracted Driving Detection: A Review of Machine Learning Techniques*. arXiv: [2505.01973](https://arxiv.org/abs/2505.01973) [cs.CV]. URL: <https://arxiv.org/abs/2505.01973> (visited on 06/22/2026).
- [6] Doshi, A. and M. M. Trivedi (2009). “On the roles of eye gaze and head dynamics in predicting driver’s intent to change lanes”. In: *IEEE Transactions on Intelligent Transportation Systems* 10.3, pp. 453–462. DOI: [10.1109/TITS.2009.2026675](https://doi.org/10.1109/TITS.2009.2026675).
- [7] Du, Y., C. Raman, A. W. Black, L.-P. Morency, and M. Eskenazi (2018). *Multimodal Polynomial Fusion for Detecting Driver Distraction*. arXiv: [1810.10565](https://arxiv.org/abs/1810.10565) [cs.HC]. URL: <https://arxiv.org/abs/1810.10565> (visited on 06/22/2026).
- [8] Elshamy, M. R., H. M. Emara, M. R. Shoaib, and A.-H. A. Badawy (2024). *P-YOLOv8: Efficient and Accurate Real-Time Detection of Distracted Driving*. arXiv: [2410.15602](https://arxiv.org/abs/2410.15602) [cs.CV]. URL: <https://arxiv.org/abs/2410.15602> (visited on 06/22/2026).
- [9] Eraqi, H. M., Y. Abouelnaga, M. H. Saad, and M. N. Moustafa (2019). “Driver Distraction Identification with an Ensemble of Convolutional Neural Networks”. In: *Journal of Advanced Transportation* 2019, p. 4125865. DOI: [10.1155/2019/4125865](https://doi.org/10.1155/2019/4125865).
- [10] European Commission (2024). *Commission Implementing Regulation (EU) 2024/1721*. URL: https://eur-lex.europa.eu/eli/reg_impl/2024/1721/oj/eng (visited on 06/15/2026).
- [11] European Road Safety Observatory (2022). *Road Safety Thematic Report: Driver Distraction*. Tech. rep. Brussels: European Commission.
- [12] Gumaiei, A., M. Al-Rakhamei, M. M. Hassan, A. Alamri, M. Alhussein, M. A. Razzaque, and G. Fortino (2020). “A deep learning-based driver distraction identification framework over edge cloud”. In: *Neural Computing and Applications*. DOI: [10.1007/s00521-020-05328-1](https://doi.org/10.1007/s00521-020-05328-1).
- [13] Hariharan, J., R. Rama Varior, and S. Karunakaran (2023). *Real-time Driver Monitoring Systems on Edge AI Device*. arXiv: [2304.01555](https://arxiv.org/abs/2304.01555) [cs.CV]. URL: <https://arxiv.org/abs/2304.01555> (visited on 06/23/2026).
- [14] Heenetimulla, H. D. S. M., S. K. Das, G. K. Sahoo, and P. Singh (2022). “An FPGA System for Driver Mobile Usage Detection and Heart Rate-Based Health Monitoring”. In: *2022 2nd Odisha International Conference on Electrical Power Engineering, Communication and Computing Technology (ODICON)*, pp. 1–6. DOI: [10.1109/ODICON54453.2022.10010243](https://doi.org/10.1109/ODICON54453.2022.10010243).
- [15] Horgan, J., C. Hughes, J. McDonald, and S. Yogamani (2021). *Vision-Based Driver Assistance Systems: Survey, Taxonomy and Advances*. arXiv: [2104.12583](https://arxiv.org/abs/2104.12583) [cs.CV]. URL: <https://arxiv.org/abs/2104.12583> (visited on 06/22/2026).

- [16] Hossain, M. U., M. A. Rahman, M. M. Islam, A. Akhter, M. A. Uddin, and B. K. Paul (2022). “Automatic driver distraction detection using deep convolutional neural networks”. In: *Intelligent Systems with Applications* 14, p. 200075. DOI: [10.1016/j.iswa.2022.200075](https://doi.org/10.1016/j.iswa.2022.200075).
- [17] Hssayeni, M. D., S. Saxena, R. Ptucha, and A. Savakis (2017). “Distracted Driver Detection: Deep Learning vs Handcrafted Features”. In: *Electronic Imaging*. DOI: [10.2352/ISSN.2470-1173.2017.10.IMAWM-162](https://doi.org/10.2352/ISSN.2470-1173.2017.10.IMAWM-162).
- [18] Huang, Y., X. Wang, and X. Zhang (2020). “HCF: A Hybrid CNN Framework for Behavior Detection of Distracted Drivers”. In: *IEEE Access* 8, pp. 109335–109349. DOI: [10.1109/ACCESS.2020.3001159](https://doi.org/10.1109/ACCESS.2020.3001159).
- [19] Hussien, M., M. Khalil, and H. Abbas (2020). *Novel Driver Distractions Dataset With Low Lighting Support*. Version 2. Mendeley Data. DOI: [10.17632/ykmr99nrsg.2](https://doi.org/10.17632/ykmr99nrsg.2). URL: <https://data.mendeley.com/datasets/ykmr99nrsg/2> (visited on 06/23/2026).
- [20] Kaggle (2016). *State Farm Distracted Driver Detection*. URL: <https://www.kaggle.com/competitions/state-farm-distracted-driver-detection/data> (visited on 06/15/2026).
- [21] Kanimozhi, S., S. Rajalakshmi, K. S. Thanushree, and L. Varshini (2025). “Autonomous Driver Distraction Identification using the YOLO Algorithm in Deep Learning”. In: *2025 International Conference on Machine Learning and Autonomous Systems (ICMLAS)*, pp. 1710–1714. DOI: [10.1109/ICMLAS64557.2025.10968649](https://doi.org/10.1109/ICMLAS64557.2025.10968649).
- [22] Kaplan, S., M. A. Guvensan, A. G. Yavuz, and Y. Karalurt (2015). “Driver Behavior Analysis for Safe Driving: A Survey”. In: *IEEE Transactions on Intelligent Transportation Systems* 16.6, pp. 3017–3032. DOI: [10.1109/TITS.2015.2462084](https://doi.org/10.1109/TITS.2015.2462084).
- [23] Kashevnik, A., R. Shchedrin, C. Kaiser, and A. Stocker (2021). “Driver Distraction Detection Methods: A Literature Review and Framework”. In: *IEEE Access* 9, pp. 60063–60076. DOI: [10.1109/ACCESS.2021.3073599](https://doi.org/10.1109/ACCESS.2021.3073599).
- [24] Khan, M. A., J.-G. Choi, and B.-G. Lee (2023). “EFFNet-CA: An Efficient Driver Distraction Detection Based on Multiscale Features Extractions and Channel Attention Mechanism”. In: *Sensors* 23.8, p. 3835. DOI: [10.3390/s23083835](https://doi.org/10.3390/s23083835).
- [25] Kidu, W. et al. (2024). “An Intelligent Real-Time Driver Activity Recognition System Using RGBDT Sensor Module”. In: *Applied Sciences* 14.17, p. 7985. DOI: [10.3390/app14177985](https://doi.org/10.3390/app14177985).
- [26] Kopuklu, O., T. Ledwon, Y. Rong, N. Kose, and G. Rigoll (2020). *DriverMHG: A Multi-Modal Dataset for Dynamic Recognition of Driver Micro Hand Gestures and a Real-Time Recognition Framework*. arXiv: [2003.00951](https://arxiv.org/abs/2003.00951) [cs.CV]. URL: <https://arxiv.org/abs/2003.00951> (visited on 06/23/2026).
- [27] Kurian, A. T. and P. K. Soori (2023). “AI-Based Driver Drowsiness and Distraction Detection in Real-Time”. In: *2023 International Conference on Computational Intelligence and Knowledge Economy (ICCIKE)*, pp. 13–18. DOI: [10.1109/ICCIKE58312.2023.10131730](https://doi.org/10.1109/ICCIKE58312.2023.10131730).
- [28] Li, X., D. Nie, J. Yu, H. Zhang, W. Qi, Y. Hu, Y. Liu, Q. Wang, and J. Li (2022). “Driver Distraction Detection Using Octave-Like Convolutional Neural Network”. In: *IEEE Transactions on Intelligent Transportation Systems* 23.10, pp. 18499–18510. DOI: [10.1109/TITS.2021.3086411](https://doi.org/10.1109/TITS.2021.3086411).
- [29] Majdi, M. S., S. Ram, J. T. Gill, and J. J. Rodriguez (2020). *Drive-Net: Convolutional Network for Driver Distraction Detection*. arXiv: [2006.12586](https://arxiv.org/abs/2006.12586) [cs.CV]. URL: <https://arxiv.org/abs/2006.12586> (visited on 06/22/2026).
- [30] Mantrala, S., D. Valluru, and P. Nukala (2025). “FocusDrive: Edge AI Device System for Real-Time Driver Distraction Detection Using Geometric Rules and Quantum-Optimized CNNs”. In: *2025 IEEE MIT Undergraduate Research Technology Conference (URTC)*, pp. 1–5. DOI: [10.1109/URTC68753.2025.11533017](https://doi.org/10.1109/URTC68753.2025.11533017).
- [31] National Highway Traffic Safety Administration (2024). *NHTSA releases 2022 traffic deaths, 2023 early estimates*. URL: <https://www.nhtsa.gov/press-releases/2022-traffic-deaths-2023-early-estimates> (visited on 06/15/2026).

- [32] Ortega, J. D., N. Kose, P. Cañas, M.-A. Chao, A. Unnervik, M. Nieto, O. Otaegui, and L. Salgado (2020). *DMD: A Large-Scale Multi-Modal Driver Monitoring Dataset for Attention and Alertness Analysis*. arXiv: 2008.12085 [cs.CV]. URL: <https://arxiv.org/abs/2008.12085> (visited on 06/23/2026).
- [33] Page, M. J., J. E. McKenzie, P. M. Bossuyt, I. Boutron, T. C. Hoffmann, C. D. Mulrow, L. Shamseer, J. M. Tetzlaff, E. A. Akl, S. E. Brennan, R. Chou, J. Glanville, J. M. Grimshaw, A. Hróbjartsson, M. M. Lalu, T. Li, E. W. Loder, E. Mayo-Wilson, S. McDonald, L. A. McGuinness, L. A. Stewart, J. Thomas, A. C. Tricco, V. A. Welch, P. Whiting, and D. Moher (2021). “The PRISMA 2020 statement: an updated guideline for reporting systematic reviews”. In: *BMJ* 372, n71. DOI: 10.1136/bmj.n71.
- [34] PRISMA (2020). *PRISMA 2020 flow diagram*. URL: <https://www.prisma-statement.org/prisma-2020-flow-diagram> (visited on 06/15/2026).
- [35] Rahman, M. S., J. Wang, S. V. Gursoy, D. Anastasiu, S. Wang, and A. Sharma (2022). *Synthetic Distracted Driving (SynDD2) Dataset for Analyzing Distracted Behaviors and Various Gaze Zones of a Driver*. arXiv: 2204.08096 [cs.CV]. URL: <https://arxiv.org/abs/2204.08096> (visited on 06/23/2026).
- [36] Ramzan, M., H. U. Khan, S. M. Awan, A. Ismail, M. Ilyas, and A. Mahmood (2019). “A Survey on State-of-the-Art Drowsiness Detection Techniques”. In: *IEEE Access* 7, pp. 61904–61919. DOI: 10.1109/ACCESS.2019.2914373.
- [37] Regan, M. A., C. Hallett, and C. P. Gordon (2011). “Driver distraction and driver inattention: Definition, relationship and taxonomy”. In: *Accident Analysis & Prevention* 43.5, pp. 1771–1781. DOI: 10.1016/j.aap.2011.04.008.
- [38] Revathi, K. L., S. P. Ahmad, and G. Surendra (2025). “Real-Time Driver Distraction Detection via CNN Feature Extraction and PCA Regularized SVM”. In: *2025 2nd International Conference on Software, Systems and Information Technology (SSITCON)*, pp. 1–6. DOI: 10.1109/SSITCON66133.2025.11342156.
- [39] Saad, M. H., M. I. Khalil, and H. M. Abbas (2020). “End-To-End Driver Distraction Recognition Using Novel Low Lighting Support Dataset”. In: *2020 15th International Conference on Computer Engineering and Systems (ICCES)*, pp. 1–6. DOI: 10.1109/ICCES51560.2020.9334619.
- [40] Sahayadhas, A., K. Sundaraj, and M. Murugappan (2012). “Detecting driver drowsiness based on sensors: A review”. In: *Sensors* 12.12, pp. 16937–16953. DOI: 10.3390/s121216937.
- [41] Sajid, M., S. Khan, T. Hussain, K. Muhammad, A. Ullah, and S. W. Baik (2021). “An Efficient Deep Learning Framework for Distracted Driver Detection”. In: *IEEE Access* 9, pp. 169270–169280. DOI: 10.1109/ACCESS.2021.3138137.
- [42] Shen, C.-H. and S.-H. Lin (2023). “Driver Distraction Detection for Daytime and Nighttime with Unpaired Visible and Infrared Image Translation”. In: *2023 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. DOI: 10.1109/IROS55552.2023.10342206. URL: <https://ieeexplore.ieee.org/document/10342206> (visited on 06/23/2026).
- [43] Sikander, G. and S. Anwar (2019). “Driver Fatigue Detection Systems: A Review”. In: *IEEE Transactions on Intelligent Transportation Systems* 20.6, pp. 2339–2352. DOI: 10.1109/TITS.2018.2868499.
- [44] Streiffer, C., R. Raghavendra, T. Benson, and M. Srivatsa (2017). “DarNet: A Deep Learning Solution for Distracted Driving Detection”. In: *Proceedings of the 18th ACM/IFIP/USENIX Middleware Conference: Industrial Track*, pp. 22–28. DOI: 10.1145/3154448.3154452.
- [45] Tan, C. W., M. Salehi, and G. Mackellar (2020). *Detecting Driver’s Distraction using Long-term Recurrent Convolutional Network*. arXiv: 2004.11839 [cs.LG]. URL: <https://arxiv.org/abs/2004.11839> (visited on 06/22/2026).
- [46] Vicente, F., Z. Huang, X. Xiong, F. De la Torre, W. Zhang, and D. Levi (2015). “Driver gaze tracking and eyes off the road detection system”. In: *IEEE Transactions on Intelligent Transportation Systems* 16.4, pp. 2014–2027. DOI: 10.1109/TITS.2015.2396031.

- [47] Vicomtech (2020). *DMD: Driving Monitoring Dataset*. URL: <https://dmd.vicomtech.org/> (visited on 06/23/2026).
- [48] Wang, J. and Z. Wu (2022). “Model Lightweighting for Real-time Distraction Detection on Resource-Limited Devices”. In: *Computational Intelligence and Neuroscience 2022*, p. 7360170. DOI: [10.1155/2022/7360170](https://doi.org/10.1155/2022/7360170).
- [49] Wang, Y. et al. (2023). *100-Driver: A Large-Scale, Diverse Dataset for Distracted Driver Classification*. DOI: [10.5281/zenodo.8335433](https://doi.org/10.5281/zenodo.8335433). URL: <https://zenodo.org/records/8335433> (visited on 06/23/2026).
- [50] Wu, Y., Y. Yin, and J. Wu (2024). “Lightweight Deep Driver Distraction Recognition Network”. In: *2024 7th International Conference on Information Communication and Signal Processing (ICICSP)*, pp. 1040–1044. DOI: [10.1109/ICICSP62589.2024.10809215](https://doi.org/10.1109/ICICSP62589.2024.10809215).
- [51] Young, K., M. Regan, and M. Hammer (2003). *Driver Distraction: A Review of the Literature*. Tech. rep. 206. Melbourne, Australia: Monash University Accident Research Centre.
- [52] Yu, Z., K. Chen, S. Li, B. Han, C. H. Liu, and S. Wang (2022). *ROMA: Cross-Domain Region Similarity Matching for Unpaired Nighttime Infrared to Daytime Visible Video Translation*. arXiv: [2204.12367](https://arxiv.org/abs/2204.12367) [cs.CV]. URL: <https://arxiv.org/abs/2204.12367> (visited on 06/23/2026).
- [53] Zhang, S., N. Yang, and C. Lv (2024). “Highly Discriminative Driver Distraction Detection Method Based on Swin Transformer”. In: *Vehicles* 6.1, pp. 78–92. DOI: [10.3390/vehicles6010006](https://doi.org/10.3390/vehicles6010006).
- [54] Zhao, C. H., B. L. Zhang, J. He, and J. Lian (2012). “Recognition of driving postures by contourlet transform and random forests”. In: *IET Intelligent Transport Systems*. Vol. 6. 2, pp. 161–168. DOI: [10.1049/iet-its.2011.0101](https://doi.org/10.1049/iet-its.2011.0101).